



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2(Paper-3)

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5. Switch

- A switch statement in C++ is a decision-making control structure that lets you choose one block of code to execute from multiple options based on the value of an expression.
- Syntax :

```
Switch (expression)
```

```
{
```

```
Case value1:
```

```
    Statement;
```

```
    break;
```

```
case value2:
```

```
    statement;
```

```
    break;
```

```
.....
```

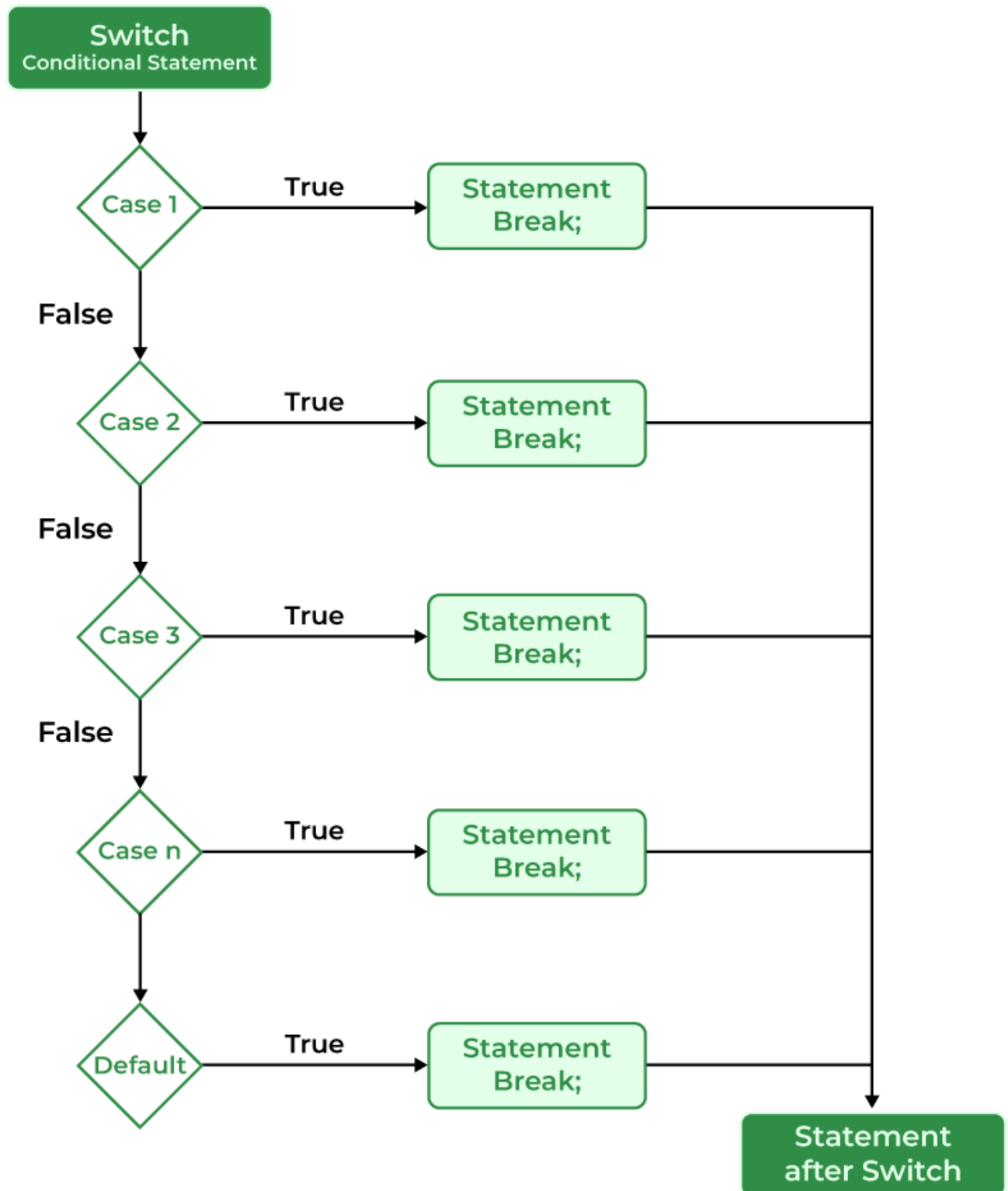
```
.....
```

```
default:
```

```
    statement;
```

```
}
```

Flowchart



Example : Write a C++ program to check odd or even.

```

#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
    int a;
    cout << " Enter any number check Odd or Even : ";
    cin >> a;
    switch(a%2)
    {
        case 0:

            cout << " EVEN ";
            break;

        case 1:

            cout << " ODD ";
            break;

        default :
            cout << " invalid";
    }
    getch();
}

```

C:\Users\hira\Desktop\hello.exe

Enter any number check Odd or Even : 50
EVEN

Question

1. Write a C++ program using switch.....case to input day number of day in a week (1 to 7) and print the day. Day number 1 means Monday, 2 is Tuesday and so on, 7 is Sunday. If the number entered is outside the range 1 to 7 then print "wrong input".
2. Write a program using switch....case to input month number (1-12) and print number of day it contains.
3. Write a program using switch.....case to print factorial or table of a number, depending upon the choice of user.
4. Write a program using switch.....case to print the gifts a buer will get when purchasing good from a super market, according to scheme given:

Purchase up to	Gift
Rs 10000	MP3 player
Rs 50000	Mp3 player + Pen drive
RS 100000	Mp3 player + pen drive + HDD
Above Rs 100000	Mp3 player + pen drive + HDD + mobile phone

5. Write a C++ program using switch statement to implement a lottery system.

6. Write a C++ program using switch.....case statement to take any number through user and print this number is +ve, -ve and 0 .
7. Write a C++ program using switch.....case statement to make calculator (add,sub,mult,divi,modu,percentage).

❖ Difference b/w if.....else and switch.....case

If.....else	switch.....case
If.....else is used to conditionally execute instructions. It provides two branching options. One when the condition is true and other when the condition is false.	switch.....case also helps in executing statements conditionally but provides multiple branching options. We can have any number of cases and each case is a branch.
If.....else can compare two values, variables or even expressions with each other.	switch.....case can only compare one variable/expression with multiple values that has to be a constant.
If.....else conditions can be a combination of various data types.	A switch.....case statement provides more clarity in comparison to equivalent if.....else statement.
An if.....else statement provides less clarity in comparison to equivalent switch.....case.	A switch.....case statement provides more clarity in comparison to equivalent if.....else statement.

Looping / Iteration Statements

Iteration (Looping) in C++

- In computer programming, loops are used to repeat a block of code
- A loop statement allows us to execute a statement or group of statements multiple times and following is the general form of a loop statement in most of the programming languages.
- For example, let's say we want to show a message 100 times. Then instead of writing the print statement 100 times, we can use a loop.

There are 3 types of loops in C++

- 1) for loop
- 2) while loop
- 3) do...while loop

1. for Loop

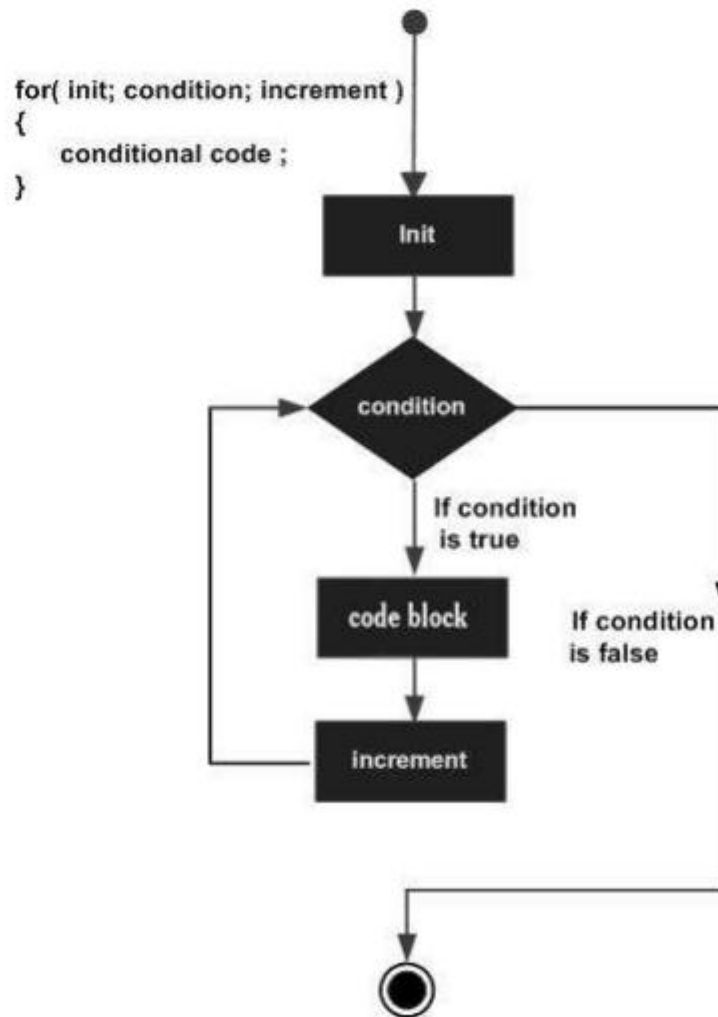
- A for loop is a repetition control structure that allows you to efficiently execute a block of code a specific number of times.
- Syntax :

```
for ( init; condition; increment )  
{  
    statement(s);  
}
```

- Example :

```
for (int i = 1; i <= 5; i++)  
{  
    cout << i << endl;  
}
```

Flow Diagram



Questions:

Q. Write a program to print your name 10 times.

Q1. Write a program to print numbers from 1 to 100 using a for loop.

Q2. Write a program to print all even numbers b/w 1 to 50 using a for loop.

Q3. Write a program to calculate and display the sum of first N natural numbers using a for loop.

Q4. Write a program to print the multiplication table of a given number using a for loop.

Q5. Write a program to find the factorial of a number using a for loop.

Q6. Write a program to count how many digits are present in a given number using a for loop.

Q7. Write a program to reverse a given number using a for loop.

Q8. Write a program to print the Fibonacci series up to N terms using a for loop.

What is Fibonacci Series → The Fibonacci series is a sequence of number where each number is the sum of the two previous numbers.

The Fibonacci Sequence

1,1,2,3,5,8,13,21,34,55,89,144,233,377...

$$1+1=2$$

$$1+2=3$$

$$2+3=5$$

$$3+5=8$$

$$5+8=13$$

$$8+13=21$$

$$13+21=34$$

$$21+34=55$$

$$34+55=89$$

$$55+89=144$$

$$89+144=233$$

$$144+233=377$$

Q9. Write a program to check whether a number is prime or not using a for loop.

Prime Numbers to 100

A prime number can only be divided (without a remainder) by itself and 1.

2	3	5	7	11
13	17	19	23	29
31	37	41	43	47
53	59	61	67	71
73	79	83	89	97

sciencenotes.org

Q10. Write a program to find the sum of digits of a number using a for loop.

Q11. write a program to check whether a number is Armstrong number or not.

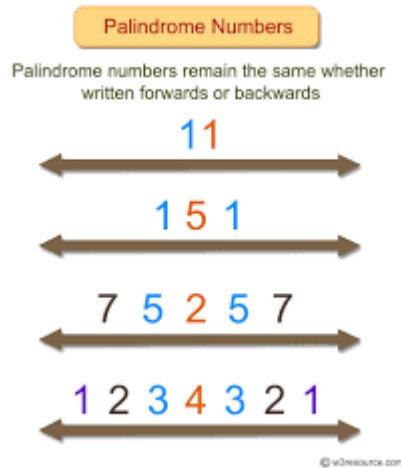
Armstrong = The sum of each digit raised to the power of the total number of digit is equal to the original number.

$$153 = 1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$$

$$371 = 3^3 + 7^3 + 1^3 = 27 + 343 + 1 = 371$$

$$9474 = 9^4 + 4^4 + 7^4 + 4^4 = 6561 + 256 + 2401 + 256 = 9474$$

Q12. Write a program to check whether a number is palindrome or not using for loop.



Q13. All pyramid patterns print using C++ program.

Q1.	Q2.	Q3.	Q4.	Q5.	Q6.
1 2 3 4 5	1 2 3 4 5	* * * * *	1 1 1 1 1	1 1 1 1 1	1 1 1 1 1
1 2 3 4 5	1 2 3 4 5	* * * * *	1 1 1 1 1	2 2 2 2 2	2 2 2 2 2
1 2 3 4 5	1 2 3 4 5	* * * * *	1 1 1 1 1	3 3 3 3 3	1 1 1 1 1
1 2 3 4 5		* * * * *	1 1 1 1 1	4 4 4 4 4	2 2 2 2 2
1 2 3 4 5		* * * * *	5 5 5 5 5		1 1 1 1 1
					2 2 2 2 2
Q7.	Q8.	Q9.	Q10.	Q11.	Q12.
#####	1 2 3 4 5	1	*	1	2
*****	2 4 6 8 10	1 1	* *	2 2	2 4
#####	3 6 9 12 15	1 1 1	* * *	3 3 3	2 4 6
*****	4 8 12 16 20	1 1 1 1	* * * *	4 4 4 4	2 4 6 8
		1 1 1 1 1	* * * * *	5 5 5 5 5	2 4 6 8 10
Q13.	Q14.	Q15.	Q16.	Q17.	
A	A	a	a	hira	
AA	AB	a a	a b	hira hira	
AAA	ABC	a a a	a b c	hira hira hira	
AAAA	ABCD	a a a a	a b c d	hira hira hira hira	
AAAAA	ABCDE	a a a a a	a b c d e	hira hira hira hira hira	

Q18.					Q19.					Q20.					Q21.					Q22.				
1					1	1	1	1	1	*	*	*	*	*	5	5	5	5	5	1	2	3	4	5
2	3				1	1	1	1		*	*	*	*		4	4	4	4		6	7	8	9	
4	5	6			1	1	1			*	*	*			3	3	3			10	11	12		
7	8	9	10		1	1				*	*				2	2				13	14			
11	12	13	14	15	1					*					1					15				

Q23.					Q24.					Q25.					Q26.					Q27.					Q28.				
	1					*					1				1	1	1	1	1	*	*	*	*	*	2	4	6	8	10
	1	1				*	*				2	2			1	1	1	1		*	*	*	*			2	4	6	8
	1	1	1			*	*	*			3	3	3		1	1	1			*	*	*	*			2	4	6	
	1	1	1	1		*	*	*	*		4	4	4	4		1	1			*	*					2	4		
1	1	1	1	1	1	*	*	*	*	*	5	5	5	5	5		1			*								2	

Q29.					Q30.					Q31.					Q32.					Q33.								
1	2	3	4	5	0					1					*					1								
	6	7	8	9	1	0				2	3				*	*	*			2	2	2						
	10	11	12		1	0	1			1	2	3			*	*	*	*	*	3	3	3	3	3				
		13	14		0	1	0	1		1	2	3	1		*	*	*	*	*	4	4	4	4	4	4	4		
			15		0	1	0	1	0	2	3	1	2	3	*	*	*	*	*	5	5	5	5	5	5	5	5	5

Q23.	Q24.	Q25.	Q26.	Q27.	Q28.
1	*	1	1 1 1 1 1	* * * * *	2 4 6 8 10
1 1	* *	2 2	1 1 1 1	* * * *	2 4 6 8
1 1 1	* * *	3 3 3	1 1 1	* * *	2 4 6
1 1 1 1	* * * *	4 4 4 4	1 1	* *	2 4
1 1 1 1 1	* * * * *	5 5 5 5 5	1	*	2

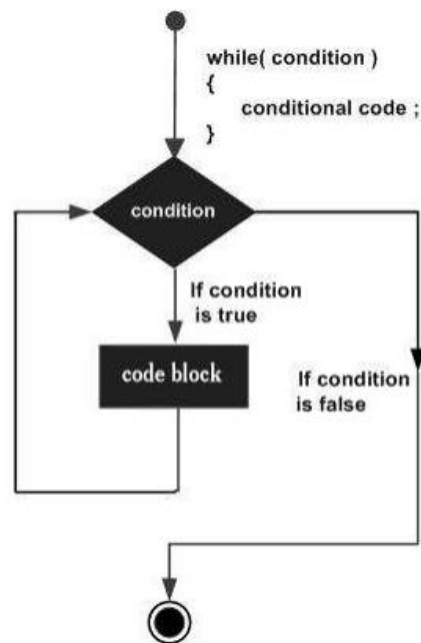
Q29.				Q30.		Q31.		Q32.				Q33.					
2	3	4	5	0		1		*				1					
6	7	8	9	10		23		***				222					
10	11	12		101		123		*****				33333					
	13	14		0101		1231		*****	*			4444444					
		15		01010		23123		*****	*	*	*	555555555					

<pre> * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * *</pre>	<pre> 1 2 2 3 3 3 4 4 4 4</pre>	<pre> 1 1 2 1 2 3 1 2 3 4</pre>	<pre> 1 2 3 4 1 2 3 1 2 1</pre>	<pre> 1 2 3 4 5 6 7 8 9 10</pre>
Square Hollow Pattern	Number Triangular	Number Increasing Pyramid	Number Increasing Reverse Pyramid	Number Changing Pyramid
<pre> 1 0 1 1 0 1 0 1 0 1</pre>	<pre> 1 2 1 2 3 2 1 2 3 4 3 2 1 2 3 4</pre>	<pre> *</pre>	<pre> * * * * * * * * * * * * * * * * * * * * * * * * *</pre>	<pre> * * * * * * * * * * * * * * *</pre>
Zero-One Triangle	Palindrome Triangular	Rhombus Pattern	Diamond Pattern	Butterfly Star Pattern
<pre> *</pre>	<pre> * * * * * * * * * * * * * * *</pre>	<pre> * * * * * * * * * * * * * * * * * * *</pre>	<pre> * * * * * * * * * * * * * * * * * * * * * * * * * * * * * *</pre>	<pre> * * * * * * * * * * * * * * * </pre>
Square Fill Pattern	Right Half Pyramid	Reverse Right Half Pyramid	Left Half Pyramid	Reverse Left Half Pyramid
<pre> +</pre>	<pre> * ** *** **** *****</pre>	<pre> 1 2 3 4 2 3 4 3 4 4</pre>	<pre> 1 2 3 4 2 3 4 3 4 4 2 3 4 1 2 3 4</pre>	<pre> * ** *** **** *****</pre>
K Pattern	Triangle Star Pattern	Reverse Number Triangle Pattern	Mirror Image Triangle Pattern	Hollow Triangle Pattern
<pre> * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * *</pre>	<pre> * ** *** **** *****</pre>	<pre> 1 2 3 4 2 3 4 3 4 4</pre>	<pre> 1 2 3 4 2 3 4 3 4 4 2 3 4 1 2 3 4</pre>	<pre> * ** *** **** *****</pre>
Hollow Reverse Triangle Pattern	Hollow Diamond Pyramid	Hollow Hourglass Pattern	Pascal's Triangle	Right Pascal's Triangle

2. While Loop

- A while loop statement repeatedly executes a target statement as long as a given condition is true.
- Syntax

```
while(condition)
{
    statement(s);
}
```



Questions

- Q1.** Write a program to print numbers from 1 to 100 using a while loop.
- Q2.** Write a program to print all odd numbers between 1 and 50 using a while loop.
- Q3.** Write a program to find the sum of first N natural numbers using a while loop.
- Q4.** Write a program to display the multiplication table of a given number using a while loop.
- Q5.** Write a program to calculate the factorial of a number using a while loop.
- Q6.** Write a program to count the digits of a given number using a while loop.
- Q7.** Write a program to reverse a number using a while loop.

Q8. Write a program to check whether a number is prime or not using a while loop.

Q9. Write a program to print Fibonacci series up to N terms using a while loop.

Q10. Write a program to find the sum of digits of a number using a while loop.

Infinite while loop case

1. `Int c=1;`
 `While (c==1)`
 {
 `Cout << "hello" << endl;`
 }
- 2 `while (5==5)`
 {
 `Cout << "hello" << endl;`
 }
3. `While (1)`
 {
 `Cout <<"hello" << endl;`
 }
4. `While (10-5)`
 {
 `Cout << "hello" << endl;`
 }

5. do...while Loop

- Unlike for and while loops, which test the loop condition at the top of the loop, the do...while loop checks its condition at the bottom of the loop.
- A do...while loop is similar to a while loop, except that a do...while loop is guaranteed to execute at least one time.
- Syntax
 `do`
 {
 `statement(s);`
 }

while(condition);

Questions

1. Write a program to print numbers form 1 to 10 using a do-while loop.
2. Write a program to print numbers form 10 to 1 in reverse order using a do-while loop.
3. Write a program to print all even numbers b/w 1 to 50 using a do-while loop.
4. Write a program to print the multiplication table of a given number using a do-while loop.
5. Write a program to find the sum of first n natural numbers using a do-while loop.
6. Write a program to count the number of digits in a given number using a do-while loop.
7. Write a program to find the reverse of a number using a do-while loop.
8. Write a program to check whether a number is palindrome or not using a do-while loop.
9. Write a program to calculate factorial of a number using a do-while loop.
10. Write a program to display a menu repeatedly until the user choose to exit (using do-while loop)

Difference between for, while, and do...while Loops in C++

Basis	for Loop	while Loop	do...while Loop
Type	Entry-controlled loop	Entry-controlled loop	Exit-controlled loop
Condition checked	Before execution	Before execution	After execution
Minimum executions	0 times	0 times	1 time
Initialization	Usually written inside the loop	Usually written before the loop	Usually written before the loop
Update	Usually written in the loop header	Usually written inside the loop body	Usually written inside the loop body
Best used when	Number of iterations is known	Number of iterations is not known in advance	The loop must execute at least once
Syntax	for(init; condition; update)	while(condition)	do { } while(condition);

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